



UC San Diego

Electrical and Computer Engineering  
JACOBS SCHOOL OF ENGINEERING

# Deep Generative Models

## Introduction and Bayesian Network

Pengtao Xie



# AI for data creation

- ❑ Images
- ❑ Texts
- ❑ Audio
- ❑ Video
- ❑ Graphs
- ❑ Computer programs
- ❑ Webpages
- ❑ Tables
- ❑ Music
- ❑ ...





# AI for data creation

ChatGPT for text generation

Stable diffusion for image generation

Sora for video generation

Music generation

Code generation



In this course, we are going to learn the methods behind these applications:

## **Deep generative models**





# Deep Generative Models

- A generative model defines a probabilistic distribution  $p(x)$  over data.
- Data generation: draw samples from  $p(x)$
- Deep generative models: use deep neural networks to define  $p(x)$
- To generate realistic data, we need to

**Define distributions  
on data**

**Incorporate  
domain knowledge**

**Understand the  
semantics of data**

**Music**

Rhythm, timbre,  
harmony, etc.

Make sense of raw  
audio signals

**Painting**

Color, tune, style,  
type, etc.

Make sense of  
raw pixels





# Deep Generative Models

Define distributions  
on data

Incorporate  
domain knowledge

Understand the  
semantics of data

Probability theory

Graph theory

Deep learning

Probabilistic graphical  
models

Deep generative  
models





# Syllabus

<https://pengtaoxie.github.io/dgm-v3.html>

New this year: Implementation of DGMs

- Over the past five years of teaching this course, most students have had little difficulty understanding the principles of DGMs or designing DGMs.
- However, many students have found it challenging to implement DGMs effectively and make them work in practice.
- To address this gap, we will devote significant instructional effort this year to teaching students how to implement DGMs and successfully make them work in real-world applications.





# Outline

- ▣ Course logistics
- ▣ Introduction







# Instructors and TAs

Instructor: Pengtao Xie, Associate Professor in ECE, [p1xie@ucsd.edu](mailto:p1xie@ucsd.edu)

Office hour: 3:30-4:30 pm Thu, Jacobs Hall 6504

TAs:

Youwei Liang, PhD student in ECE, [yol049@ucsd.edu](mailto:yol049@ucsd.edu)

Office hour: 5-6 pm Tue, Jacobs Hall 5101D

Li Zhang, PhD student in ECE, [liz042@ucsd.edu](mailto:liz042@ucsd.edu)

Office hour: 3-4 pm Mon, Jacobs Hall 5101E

Ruiyi Zhang, PhD student in ECE, [ruz048@ucsd.edu](mailto:ruz048@ucsd.edu)

Office hour: 4-5pm Wed, Jacobs Hall 5101D

Qi Cao, PhD student in ECE, [q9cao@ucsd.edu](mailto:q9cao@ucsd.edu)

Office hour: 4-5 pm Fri, Jacobs Hall 4506





# Grading

## Grading structure

- 3 homework, each 10%
- Project, 70%
  - # Project proposal (10%)
  - # Mid-term report (20%)
  - # Final report (40%)





# Project

- ❑ Proposal
  - ❑ Objective and significance, background
  - ❑ 2 pages (NeurIPS format)
- ❑ Mid-term report
  - ❑ Method, algorithm, theory
  - ❑ 5 pages (including revised proposal, excluding references)
- ❑ Final report
  - ❑ Experiments, conclusions
  - ❑ 8 pages (including revised mid-term report, excluding references)





# Project

- ❑ Topics
  - ❑ Any topic relevant to deep generative models
  - ❑ Please feel free to propose your own topic
- ❑ Team
  - ❑ 1-3 students





# Late Policy

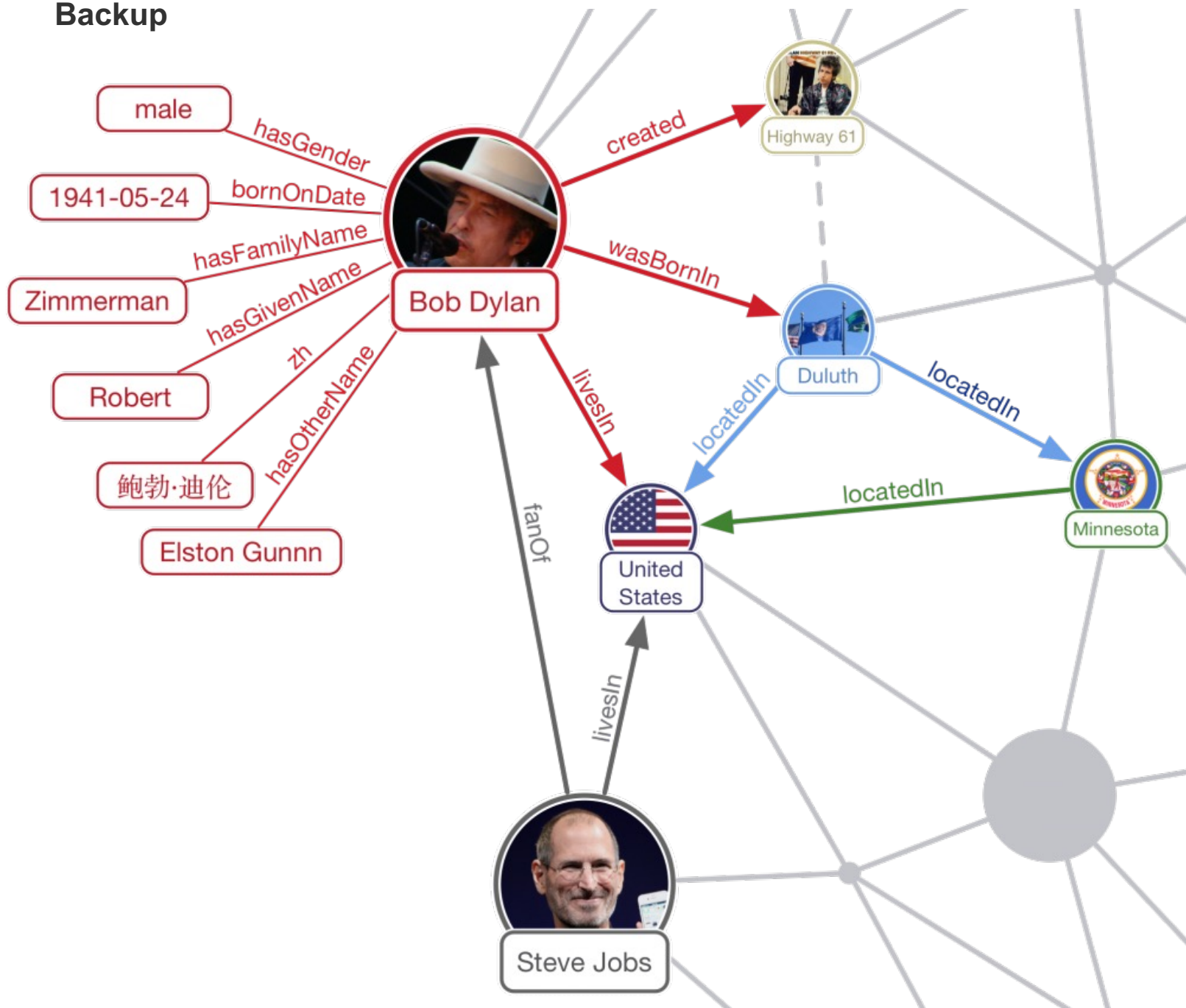
- ❑ Please submit assignments on time
- ❑ Late for 0-24 hours:  $75\% \times \text{original grade}$
- ❑ Late for 24-48 hours:  $50\% \times \text{original grade}$
- ❑ Late for  $> 48$  hours: zero
- ❑ Grace days
  - ❑ Without late penalty
  - ❑ 5 days in total for all assignments
  - ❑ Count by days, not hours
    - ❑ (0,24] hours are 1 day, (24,48] hours are 2 days, ...
  - ❑ Inform the TA that you will use X grace days for assignment Y





# Knowledge graph

Backup





# Deep Generative Models

- ❑ What deep generative models are
- ❑ Deep generative models for data understanding
- ❑ Generative models versus discriminative models
- ❑ Conditional deep generative models
- ❑ Key problems in deep generative models



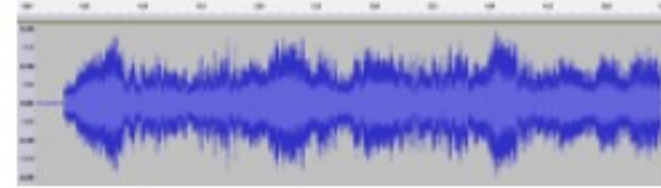


# Deep general models are not just for data generation, but also data understanding

**Challenge:** understand complex, unstructured inputs



Computer Vision



Computational Speech



Natural Language Processing



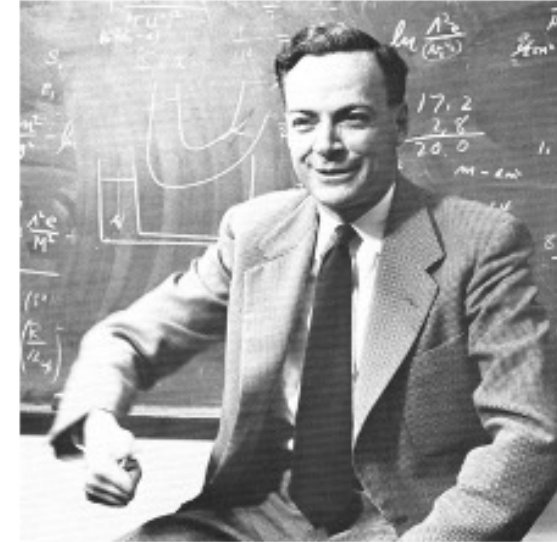
Robotics







# Deep general models are not just for data generation, but also data understanding



Richard Feynman: *"What I cannot create, I do not understand"*

Generative modeling: *"What I understand, I can **create**"*





# Generation versus understanding

How to generate natural images with a computer?

High level  
description

```
Cube(color=blue, position=, size=, ...)  
Cylinder(color=red, position=, size=,..)
```

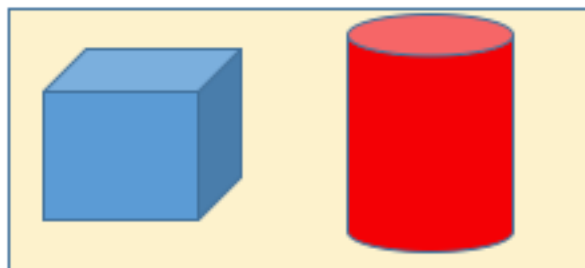
**Generation** (graphics)



**Inference** (vision)



Raw sensory  
outputs



Our models will have **similar structure (generation + inference)**





# Deep Generative Models

- ❑ What deep generative models are
- ❑ Deep generative models for data understanding
- ❑ **Generative models versus discriminative models**
- ❑ Conditional deep generative models
- ❑ Key problems in deep generative models





# Discriminative vs. generative

**Discriminative:** classify bedroom vs. dining room



The input image  $X$  is always given. **Goal:** a good decision boundary, via **conditional distribution**

$$P(Y = \text{Bedroom} \mid X = \text{[dining room image]}) = 0.0001$$

Ex: logistic regression, convolutional net, etc.

**Generative:** generate  $X$



The input  $X$  is **not** given. Requires a model of the **joint distribution**

$$P(Y = \text{Bedroom}, X = \text{[bedroom image]}) = 0.0002$$





## Discriminative vs. generative (cont'd)

Joint and conditional are related via **Bayes Rule**:

$$P(Y = \text{Bedroom} \mid X = \text{img}) = \frac{P(Y = \text{Bedroom}, X = \text{img})}{P(X = \text{img})}$$

**Discriminative:** X is always given, does not need to model

$$P(X = \text{img})$$

Therefore it cannot handle missing data

$$P(Y = \text{Bedroom} \mid X = \text{img})$$





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- ▣ Course logistics
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# Deep Generative Models

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- ❑ **Conditional deep generative models**
- ❑ Key problems in deep generative models





# Conditional Generative Models

Class **conditional generative models** are also possible:

$$P(X = \text{img} \mid Y = \text{Bedroom})$$

It's often useful to condition on rich side information Y

$$P(X = \text{img} \mid \text{Caption} = \text{"A black table with 6 chairs"})$$

A discriminative model is a very simple conditional generative model of Y:

$$P(Y = \text{Bedroom} \mid X = \text{img})$$

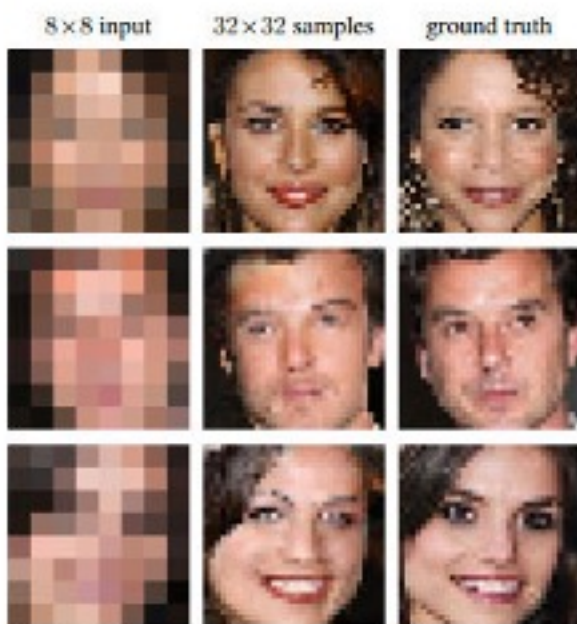






# Image Super Resolution

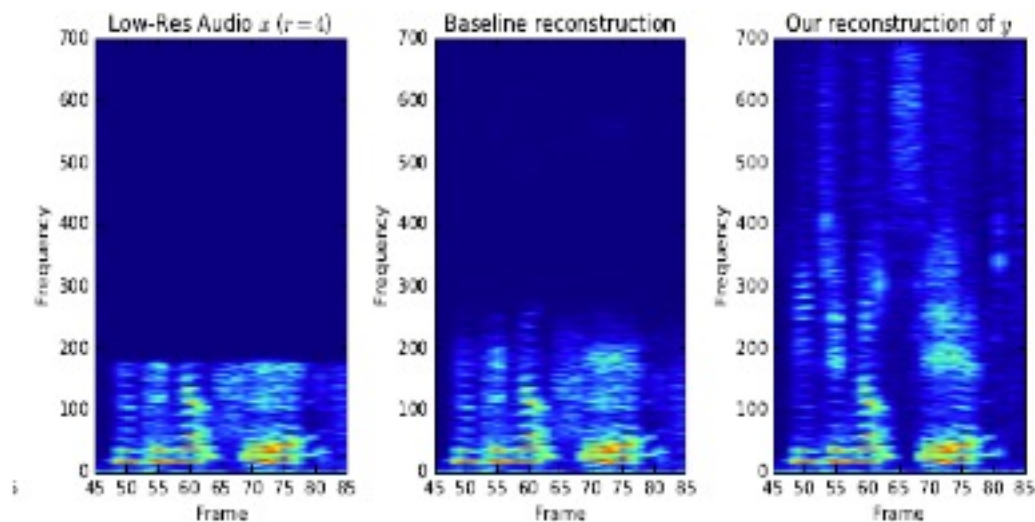
Conditional generative model  $P(\text{high res image} \mid \text{low res image})$





# Audio Super Resolution

Conditional generative model  $P(\text{high-res signal} \mid \text{low-res audio signal})$



Low res signal



High res audio signal





# Machine Translation

Conditional generative model  $P(\text{English text} | \text{Chinese text})$

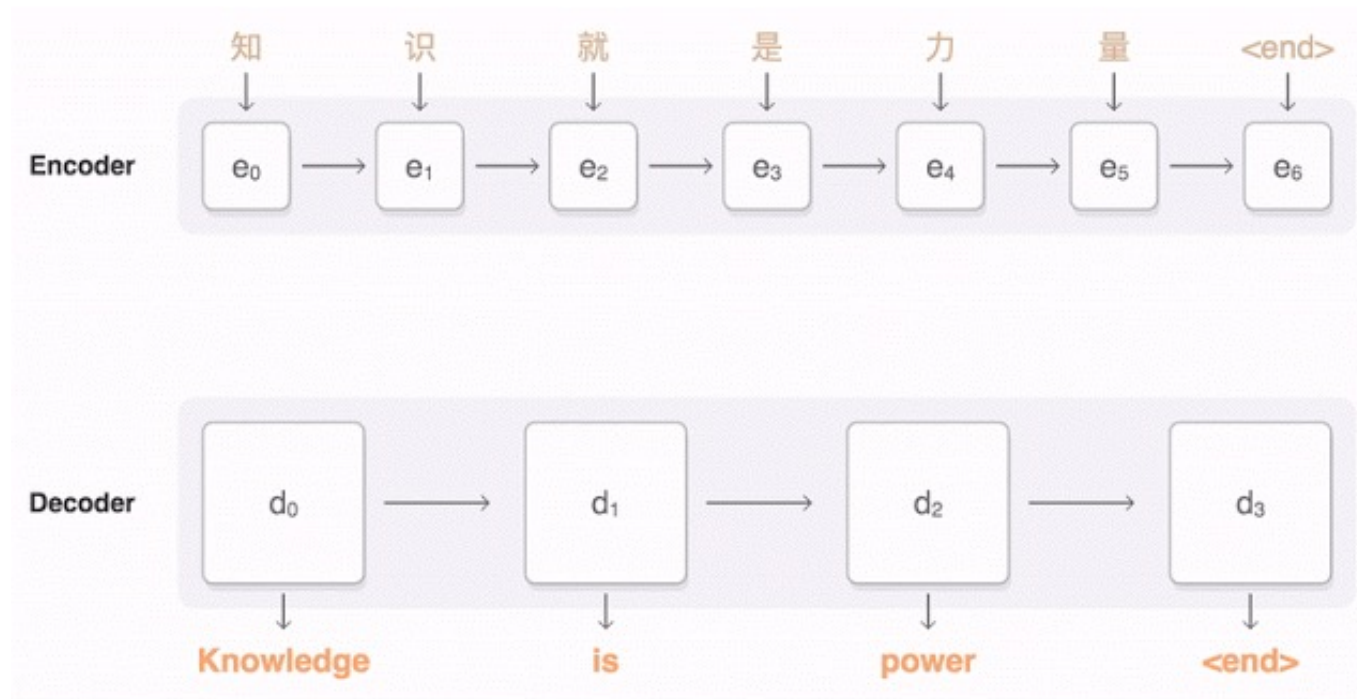


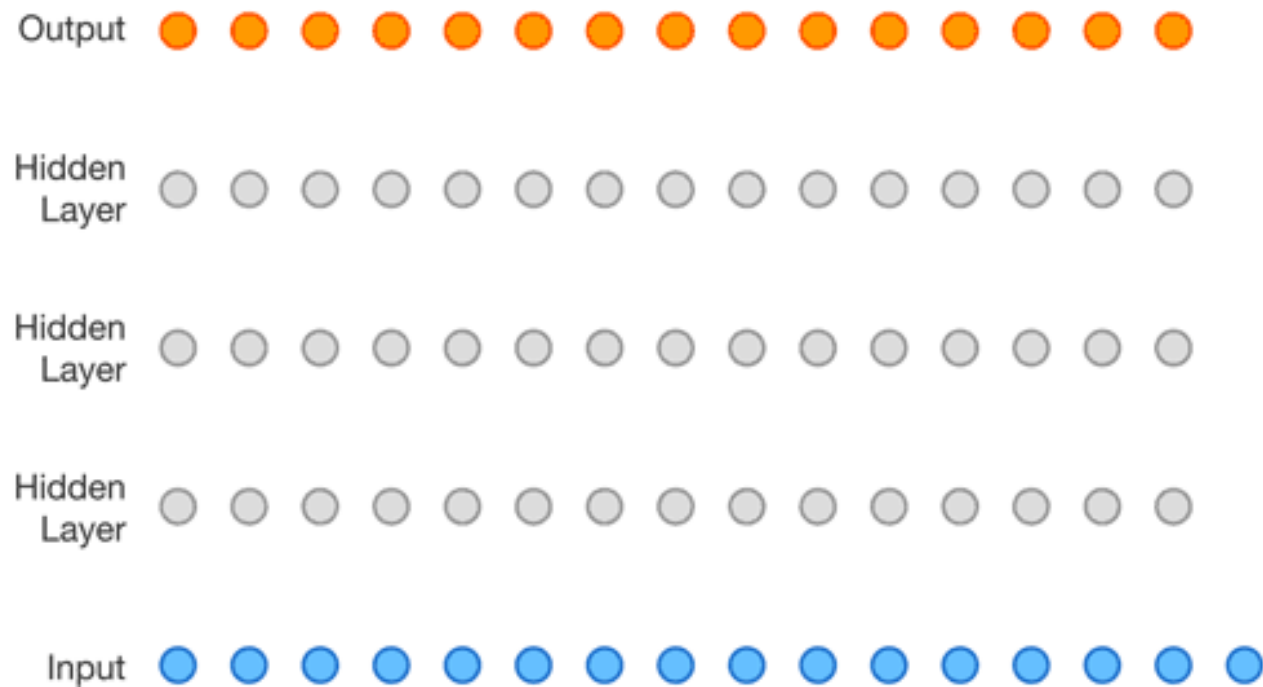
Figure from Google AI research blog.





# WaveNet

Generative model of speech signals



## Text to Speech



**Parametric**



**Concatenative**



**WaveNet**



**Unconditional**



**Music**

van den Oord et al, 2016c





# Conditional text generation

Custom prompt

To get an A+ in deep generative models, students have to

## Completion

**To get an A+ in deep generative models, students have to** be willing to work with problems that are a whole lot more interesting than, say, the ones that most students work on in class. If you're a great student, the question above can be avoided and you'll be able to do great work, but if you're not, you will need to go beyond the basics before getting good.

Now to be clear, this advice is not just for the deep-learning crowd; it is good advice for any student who is taking his or her first course in machine learning.

The key point is that if you have a deep, deep brain of a computer scientist, that's just as important to you.

**Radford et al., 2019**  
**Demo from [talktotransformer.com](http://talktotransformer.com)**







# Image Translation

Conditional generative model  $P(\text{zebra images} | \text{horse images})$



Zhu et al., 2017





# Deep Generative Models

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- ❑ Generative models versus discriminative models
- ❑ Conditional deep generative models
- ❑ **Key problems in deep generative models**



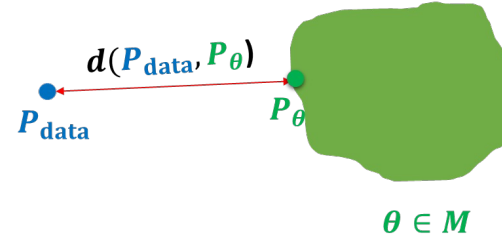


# Key problems in deep generative models

- **Representation:** how do we model the joint distribution of many random variables?
  - Need compact representation
- **Learning:** what is the right way to compare probability distributions?



$$\begin{aligned} \mathbf{x}_i &\sim P_{\text{data}} \\ i &= 1, 2, \dots, n \end{aligned}$$



**Model family**

- **Inference:** how do we invert the generation process (e.g., vision as inverse graphics)?
  - Unsupervised learning: recover high-level descriptions (features) from raw data







# Outline

- ▣ Bayesian Network
  - ▣ Directed probabilistic graphical models





# Probabilistic graphical models (PGMs)

- A probabilistic language for describing the physical world

## Physical world

**Objects** (e.g., fever, cold, etc.)

**Relations** (e.g., fever is a symptom of cold)

**Randomness** (nearly nothing is certain)

## PGMs

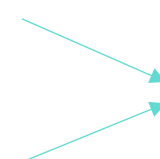
**Nodes in a graph**

**Edges in a graph**

**Random variables**

**Graph theory**

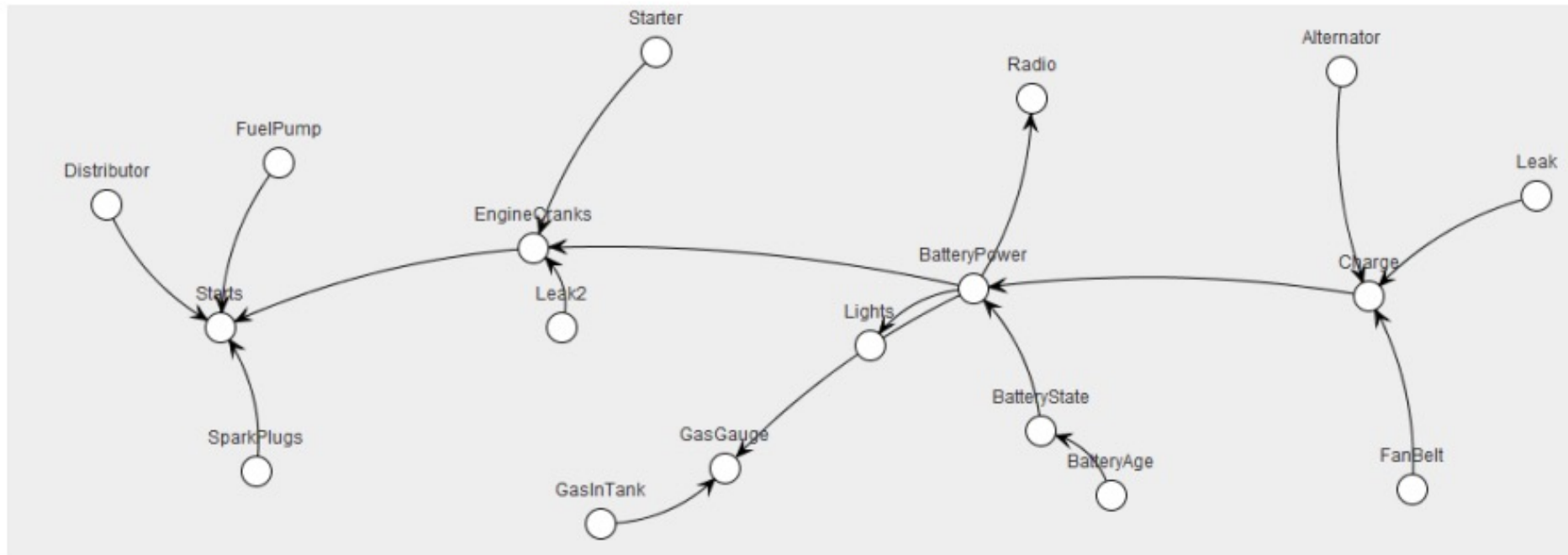
**Probability theory**





# Example 1

Will my car start this morning?

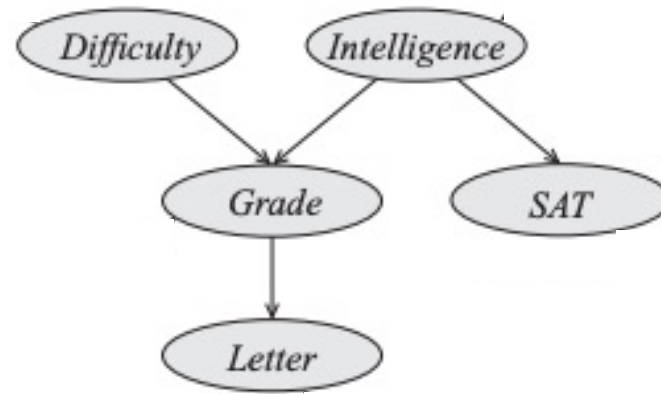


Heckerman *et al.*, Decision-Theoretic Troubleshooting, 1995





## Example 2

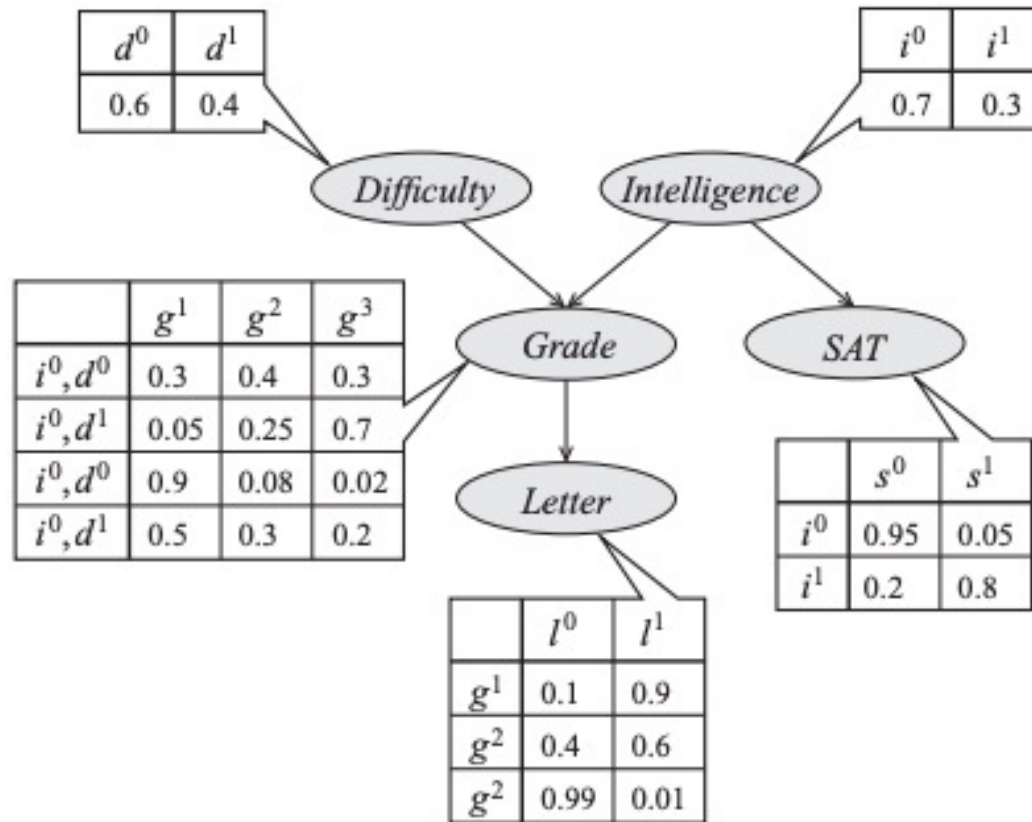


Example from Koller &  
Friedman, *Probabilistic  
Graphical Models*, 2009





## Example 2 (Cont'd)



Example from Koller & Friedman, *Probabilistic Graphical Models*, 2009





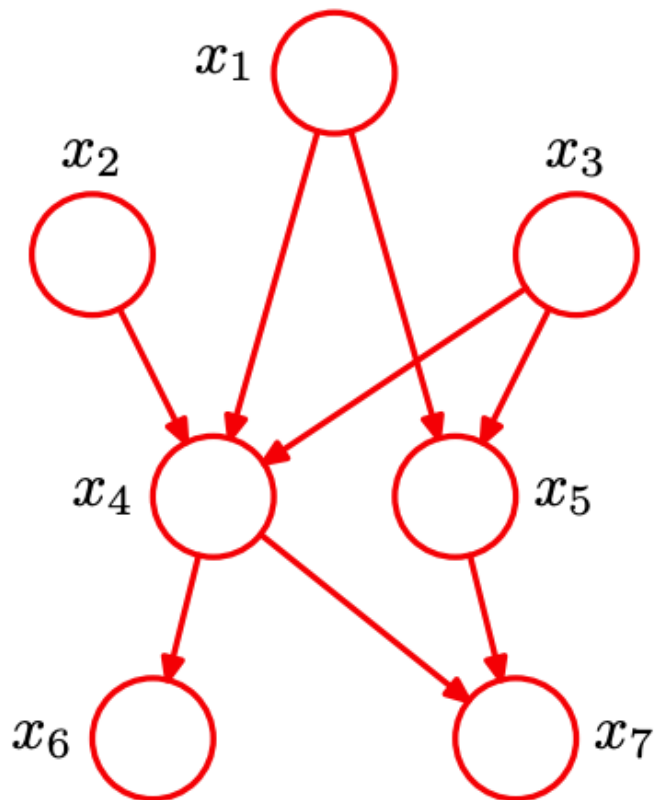
# Bayesian Network

- Bayesian network and joint distribution
- Conditional independence





# Bayesian Network



$$p(\mathbf{x}) = \prod_{k=1}^K p(x_k | \text{pa}_k)$$

$$p(x_1)p(x_2)p(x_3)p(x_4|x_1, x_2, x_3)p(x_5|x_1, x_3)p(x_6|x_4)p(x_7|x_4, x_5)$$



Mathematical foundation of DGMs

Linear algebra  
Probability  
Statistics  
Calculus  
Sampling

DGM  $\rightarrow$  Data generation  
Distribution  $\rightarrow$  Sampling

Gaussian  
Bernouli  
Dirichlet

$p(x_1)p(x_2)p(x_3)p(x_4|x_1,x_2,x_3)p(x_5|x_1,x_3)p(x_6|x_4,x_5)p(x_7|x_4,x_5,x_6)$

Classroom Technology Support  
its.ucsd.edu

Large language model

$p(w_1, w_2, w_3, w_4, \dots, w_{100})$   
categorical distributions

objects Words  
Relations I like

$p(w_k | w_{1:k-1})$   
Transformer

apple ✓  
tomorrow x  
garbage x

I like apple 0.1  
I like today 0.01

$p(w_1, w_2, \dots, w_{100})$   
 $= p(w_1) p(w_2 | w_1) p(w_3 | w_1, w_2) \dots$